Venkata Lakshmi Parimala Pasupuleti

AI/ML Backend Engineer · RAG Pipelines · LLM Workflows · Agentic Systems · LangChain · LangGraph · Transformer Models · Prompt Engineering · Python · PyTorch · Microservices · Observability

+1-703-261-5410 | venkatalakshmiparimala@gmail.com | linkedin.com/in/venkata-lakshmi-parimala-pasupuleti | github.com/venkatalakshmiparimala/AgentProbe | Virginia, USA — Open to Relocation

PROFESSIONAL SUMMARY

AI/ML backend engineer with 8 years shipping production systems and hands-on depth in **LLM workflows, RAG pipelines, transformer models, and agentic systems**. Built **Leasel**: a production RAG-based knowledge and retrieval system with an **8-agent LangGraph agentic backend** — explicit control flow, conditional edges, and state management across Triage, Retrieval, Reasoning, Validation, and HITL escalation nodes — achieving 93.6% precision on 1,240 compliance documents. Built **AgentProbe**: a scalable ML evaluation service with **latency observability, CI-integrated testing, and multi-provider LLM routing** (OpenAI, Anthropic, Groq, Ollama); validated 1,240+ test cases via pass@k harness and LLM-as-judge scoring. At Entrata: **Python microservices backend** serving 100K+ MAU at 99.5% uptime; RabbitMQ async processing; AWS data pipelines; 200+ SQL query optimizations (30% gain). M.S. Data Analytics Engineering, George Mason University (GPA 3.87).

WORK EXPERIENCE

George Mason University

Aug 2024 – May 2026

Research Associate / Graduate Teaching Assistant — Python · PyTorch · ML Pipelines · Statistical Modeling

- Built multi-source Python ML pipelines (190+ countries, WHO + UK Road Safety datasets) achieving **$R^2 = 0.977$**; applied PyTorch and Isolation Forest for model training and automated data integrity validation across 3 research pipelines. Delivered Python and ML workshops to **600+ students**.

Entrata — Enterprise PropTech SaaS

Sep 2018 – Aug 2024 · Remote

Senior Software Engineer — Python · Microservices · REST APIs · SQL · AWS · Observability

- **Python microservices backend**: 20+ production modules serving 100K+ MAU at 99.5% uptime; **cross-functional delivery** across 120+ Agile sprints; Docker microservices redesign (50% scalability improvement, 30% deployment time reduction); RabbitMQ async processing with retry logic and idempotency guarantees; Redis caching; CI/CD automation (GitHub Actions) with pre/post validation gates.
- **Observability and reliability**: built internal telemetry and debugging tooling reducing debugging time 40%; took full ownership of production incidents - RCA with corrective documentation; performance monitoring across 99.5% uptime SLA; 200+ SQL query optimizations (30% gain) via execution plan analysis, indexing, partitioning, and stored procedures.
- Led PHI-compliant **AWS S3 data pipeline migration** (Libsodium, zero security incidents); Python/Bash automation eliminating 80% of manual workload; **300+ code reviews; mentored 5+ engineers**.

Entrata India (Db Xento Systems)

Jan 2017 – Sep 2018 · Pune

Associate / Trainee Software Engineer — Python · SQL · REST APIs · AWS

- Built Python and SQL backend modules; designed data integration pipelines processing 1.6M federal housing records; led AWS S3 cloud migration; contributed to API development and defect analysis.

PROJECTS

Leasel: RAG Knowledge System & 8-Agent Agentic Backend

RAG · AGENTIC

Python · LangGraph · LangChain · FastAPI · ChromaDB (Vector DB) · Transformer Models (BERT) · TF-IDF RAG · SQLite · Docker · Streamlit — github.com/venkatalakshmiparimala/Lease_complaince_rag

- **RAG retrieval and knowledge system**: section-aware chunking, TF-IDF + semantic retrieval (ChromaDB vector database), transformer-based embeddings (BERT), 100% citation correctness, missing/conflicting clause detection — validated across 1,240 compliance documents (5 US states + federal).
- **Agentic backend with explicit control flow**: 8-agent LangGraph StateGraph with conditional edges and state management (Triage → Retrieval → Configuration → Historical → Structured Data → Reasoning → Validation → HITL Escalation) — produces ranked, evidence-backed root-cause hypotheses; confidence-scored outputs trigger deterministic fallbacks or human escalation. Core design: **LLM reasons; deterministic logic makes every regulated call**.
- Results: **93.6% precision, 87.9% F1, 0% unsupported-claim rate**; pass@1 81.7%, pass@3 88.3% via pass@k harness with LLM-as-judge scoring.

AgentProbe: Scalable ML Evaluation Service — Observability · Latency · CI Testing

ML SERVICE · EVAL

Python · FastAPI · LangGraph · PyTorch · ChromaDB · React · Docker · CI/CD (GitHub Actions) — github.com/venkatalakshmiparimala/AgentProbe

- **Scalable ML service with full observability**: 15-endpoint FastAPI backend; per-run **latency tracking**, retrieval drift monitoring, and performance metrics via React dashboard; **CI-integrated testing gate** (GitHub Actions) blocking regressions before merge; multi-provider LLM routing (OpenAI, Anthropic, Groq, Ollama) with zero vendor lock-in.
- 5-metric eval engine (correctness, groundedness, hallucination rate, latency, token efficiency); **pass@k harness** with multi-trial sampling and LLM-as-judge scoring — validated across **1,240+ test cases**: pass@1 81.7%, pass@3 88.3%.

AI on FHIR: NLP Transformer Pipeline & Enterprise REST API Integration

NLP · APIS

Python · Flask · REST APIs (FHIR) · Transformer Models (BERT) · spaCy · Docker · React — github.com/venkatalakshmiparimala/AI_on_FHIR

- Python Flask microservice integrating with FHIR REST APIs — **transformer-based NLP pipeline** (BERT + spaCy + LLM) translating natural-language queries into structured API calls; HIPAA-compliant RBAC; Docker containerized deployment; ~60% reduction in manual retrieval time. Demonstrates **production-ready ML service with enterprise API integration**.

EDUCATION

M.S. Data Analytics Engineering — George Mason University — May 2026 · GPA 3.87

ML systems, NLP, LLM pipelines, RAG, statistical modeling, performance engineering, observability

B.Tech Computer Science & Engineering — JNTU Anantapur — 2013 – 2017

TECHNICAL SKILLS

Engineering & Backend: Python (expert), SQL (expert — 200+ production optimizations, schema design, execution plan tuning, indexing, partitioning, stored procedures, triggers), PL/SQL, PHP, JavaScript, TypeScript, R; FastAPI, REST APIs, Microservices, RabbitMQ, Redis (caching), Docker, CI/CD (GitHub Actions), Deployment Automation, AWS (S3, EC2, RDS, Glue), PostgreSQL, MySQL, Oracle, SQLite, Git, Linux

AI & Agentic: LangChain, LangGraph, Agentic Workflows, Control Flow & State Management, RAG Pipelines, Retrieval Systems, Vector Databases (ChromaDB), LLM Integration (OpenAI · Anthropic · Groq · Ollama), Transformer Models, Embeddings, Semantic Search, Prompt Engineering (few-shot, chain-of-thought, output validation, context engineering), HITL, Confidence Scoring, Guardrails, Responsible AI, Streamlit, React, NLP (spaCy, BERT), MCP

ML & Analytics: PyTorch, scikit-learn, XGBoost, Random Forest, Isolation Forest, Feature Engineering, Anomaly Detection, Statistical Modeling, Model Evaluation, Tableau, Power BI, ggplot2, Agile/Scrum, Planning Poker; 300+ code reviews, 5+ engineers mentored